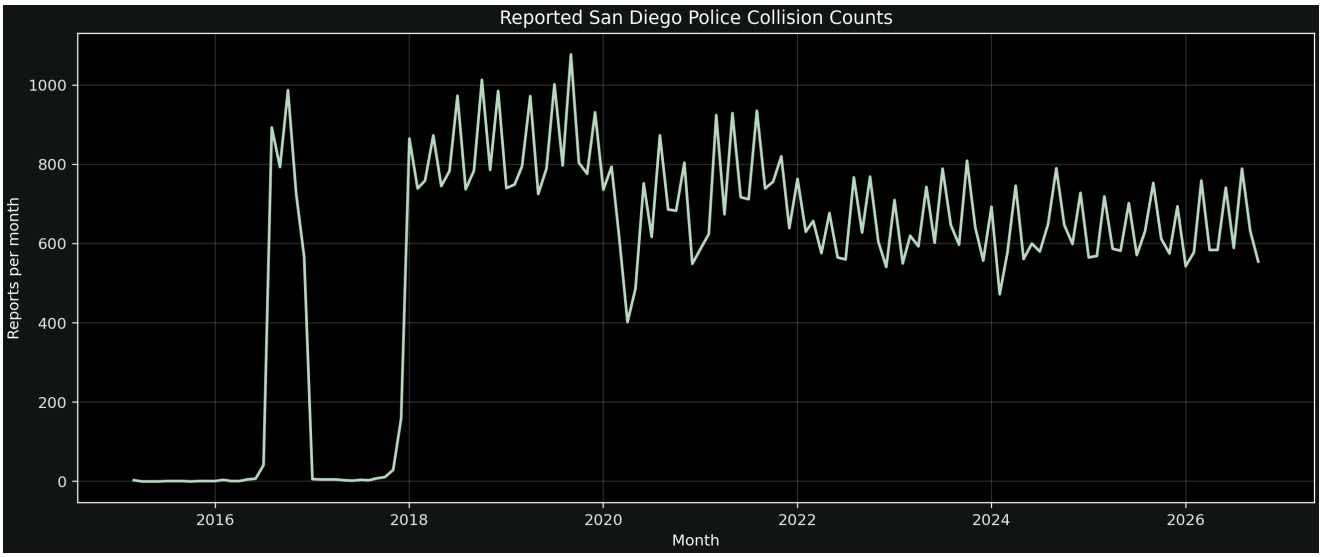
CRASHCAST

A reproducible framework for weekly collision-risk ranking in San Diego

APPLIED URBAN SAFETY ANALYTICS / RESEARCH RELEASE 0.1

Executive summary

CrashCast estimates relative collision risk for the next week across San Diego Police beats. It is an aggregate forecasting study, not individual-crash or street-level prediction. The source snapshot contains 78,715 unique police reports from 2015-03-19 to 2026-11-06 across 148 beats. On an untouched 2025 test year, the selected model achieved MAE 0.698, compared with 0.888 for a seasonal-naive baseline.



Data provenance and research scope

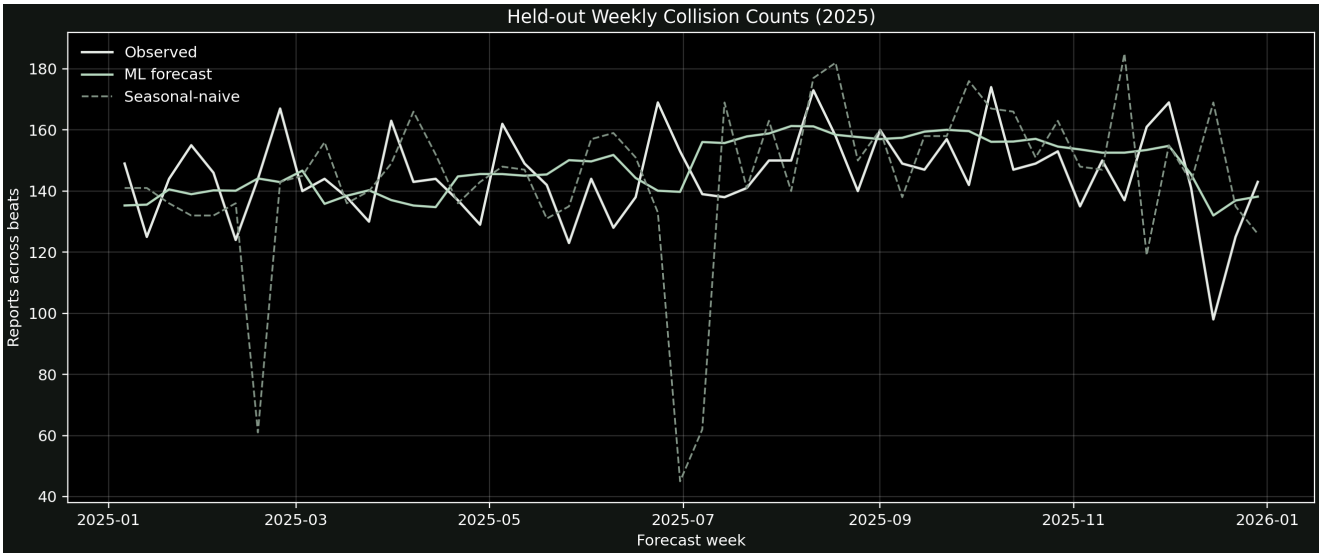
The primary source is the City of San Diego Police traffic-collision basic reports dataset. It supplies report time, police beat, injury, fatality, and hit-and-run fields. It is updated daily, excludes many property-damage-only events, and does not cover freeway crashes handled by CHP. The source has block-level address components but no verified coordinates; CrashCast therefore uses police beats rather than fabricating point locations. Selected traffic-count observations are retained in provenance but not treated as exposure without a validated spatial crosswalk.

Problem definition and feature construction

For each police beat and Monday-starting week, the target is the following week's report count. Features are one-, two-, four-, eight-, twelve-, and fifty-two-week lags; shifted rolling count, injury, fatality, and hit-and-run summaries; beat identity; and cyclical week-of-year terms. Every history feature is shifted before the forecast week. The last incomplete source week is excluded.

Validation and findings

Training uses data through 2023, model selection uses 2024, and all reported results use 2025 only (7,696 beat-week rows). The selected Poisson histogram gradient-boosting model had test RMSE 1.026 and Poisson deviance 0.935. The separate next-week-occurrence classifier had average precision 0.815 and Brier score 0.162.



Limitations and responsible use

Predictive association is not causation. Collision reporting, enforcement, road exposure, and data coverage differ across time and neighborhoods. Police beats are administrative rather than road-network areas. CrashCast is unsuitable for individual decisions, enforcement allocation, insurance pricing, or any inference about people or communities.

Reproducibility

Run PYTHONPATH=src python scripts/run_research.py after downloading the official source. The pipeline writes source hashes, validation metadata, experiment metrics, prediction rows, and figures. Raw data, row-level outputs, and models are intentionally excluded from the public repository unless their redistribution terms are reviewed.

References

City of San Diego, Traffic collisions - basic reports: https://data.sandiego.gov/datasets/traffic-collisions/
City of San Diego, Traffic collision details: https://data.sandiego.gov/datasets/traffic-collision-details/
City of San Diego, Traffic ADT counts: https://data.sandiego.gov/datasets/traffic-adt-counts/